



Exploring the Future of Prompt Engineering in Healthcare: Mission and Vision, Methods, Opportunities, Challenges, Issues and Their Remedies, Contributions, Advantages, Disadvantages, Applications, and Algorithms

Chinnem Rama Mohan¹ · Rapelli Naga Sathvik¹ · Chitta Kushal¹ · S. Kiran² · A. Ashok Kumar³

Received: 22 April 2024 / Accepted: 28 September 2024
© The Institution of Engineers (India) 2024

Abstract Prompt engineering in healthcare has appeared as a significant factor in improving the various technologies in the medical field, with a particular focus on disease diagnosis and curing. This research paper focuses on the importance of creating proper prompts that improve the efficiency of the healthcare system, especially in diseases such as cancers, including breast and lung cancers, heart-related issues, and various other diseases. This also helps healthcare experts and researchers diagnose diseases earlier, provide patient care, and select the suitable treatment after diagnosis. The research discovers various prompt generation methods and their impact on healthcare, including simple rule-based systems and machine learning algorithms. The paper also discusses various ethical considerations in prompt engineering as confidential medical information will be present, particularly in disease diagnosis. By studying several case studies and relevant literature, this paper focuses on contributing valuable data to prompt engineering in the healthcare domain. It focuses on improving the early detection and management of diseases like cancer, heart-related diseases, and various others. This work ultimately focuses on making the healthcare system more advanced and making the patient's life much better and easier.

Keywords Prompt engineering · Artificial intelligence · Large language models · Disease diagnosis · Stable diffusion · Healthcare domain · Machine learning · Deep learning · Early detection · Curing cancer

Introduction

The term "prompt engineering" was first named by Tom B. Brown et al. in their research paper "Language Models are Few-Shot Learners." They have discussed this topic to bring better results from Large Language Models (LLMs) to perform various tasks. Although prompt engineering is a new field, its traces can be found in the early days of artificial intelligence and deep learning research. Also, this is an emerging concept of natural language processing. Initially, the researchers tried to use prompt engineering to train the computer using natural language. These were so limited for interaction.

In the 1950–1960s, the researchers started the basics of prompting as they were just able to understand that they would be able to control computers using prompts. In 1970–1980, the researchers advanced their prompting techniques in such a way that using computers to generate creative content from the existing data. Between 1990 and 2000, prompt engineering started gaining more popularity as Artificial Intelligence (AI) and Machine Learning (ML) evolved. From 2015 to the present, 'Prompt Engineering' has emerged as one of the latest trends, with the birth of millions of LLMs. The prompting is crucial to make those LLMs very efficient and helpful. With better prompting techniques, the LLMs would be able to generate the exact results that were expected. Prompt engineering is continuously getting advanced, so a more significant number of prompting techniques will evolve in the future. Also, with the latest

✉ Chinnem Rama Mohan
ramamohanchinnem@gmail.com

¹ Department of CSE, Narayana Engineering College, Nellore, Andhra Pradesh 524004, India

² Department of CSE, YSR Engineering College of YVU, Proddatur, Andhra Pradesh, India

³ Department of Physics, YSR Engineering College of YVU, Proddatur, Andhra Pradesh 516360, India

tools like ChatGPT, it is essential to make sure that giving accurate prompts is important [1, 2].

These days, technology is evolving rapidly. One such evolved technology is Artificial Intelligence. AI helps machines think like humans, thus helping humans do their work faster and making life easier. To train these AI models, we use an extensive amount of data and deep learning techniques in order to make them work. These models are tested in various environments to test their efficiency and accuracy. Later, these models are released for use. In order to use these models and achieve accurate results, the user must be able to communicate with the AI model. This process of communication involves writing effective prompts [2].

A prompt is a set of instructions that consists of the task to be done by large language models. Generating a prompt requires a lot of skill, and one has to be good at the language (English) because it is the default language used everywhere, and most of the LLMs are built using the same language. An effective prompt consists of various components like Task, Context, Format, etc.

Prompt engineering is a technique of developing effective prompts that will make the AI models generate the expected results. Prompt engineering has gained sudden attention with the advancement of AI tools. In order to develop accurate prompts, it requires creating the prompt with various components. Following these components will primarily ensure that the results being generated are effective and meet the requirements.

Components of Prompts

Various components in prompting affect the responses from models. As shown in Fig. 1, there are six main components that have to be followed while giving prompts to an LLM to get better results.

Persona

This represents the personality of the AI model, which refers to how the model should behave or act when responding to that particular prompt.

Ex_Prompt1: "Assume the role of a skilled LinkedIn content creator and write an informative LinkedIn post for promoting a product."

In the above prompt, the model will behave like a skilled LinkedIn content writer. Thus, the persona/personality for the above is a 'skilled LinkedIn content creator.' Thus, the content will sound like it was written by that person.

Context

Context is also a crucial component in generating better results because it will act like the information or input data

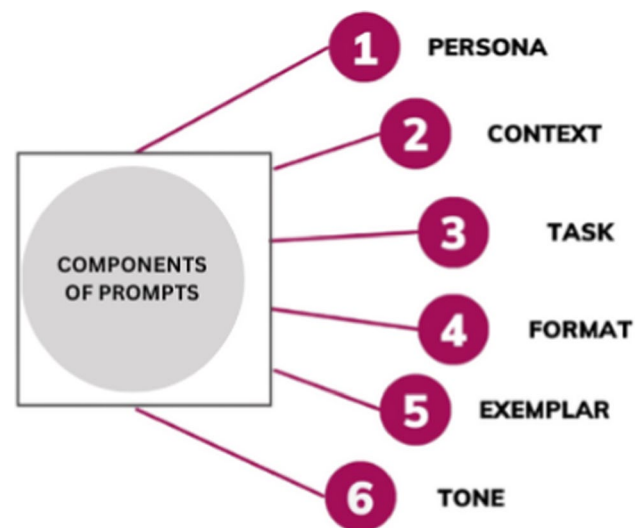


Fig. 1 Components of prompts

for the prompt. Thus, the more context you provide, the better the result will be.

Ex_Prompt2: "You are a senior manager at Google and you have just released a new product which is receiving large number preorders. Now your task is to write an email to your boss sharing the positive news."

In the above example, the context informs the model about the "product that has been successful." This information will be useful for the AI model to generate quality output.

Task

This is the most critical component. It suggests the model of the actual action that has to be performed. Let us consider the previous example. In the prompt, "writing an Email," is the task that the model will perform.

Format

Format refers to how the result has to be generated. Considering the above example, the format of the result will be the "Email." These models are trained in such a way that they will be able to generate any results when a proper prompt is provided.

Exemplar

Exemplar refers to providing extra constraints for the format, like what aspects should be addressed while generating the result. Here is a continuation of the above prompt.

Ex_Prompt3: "The email should include a tl;dr (too long, didn't read) section, project background (why this product

has come into existence), and end with a section thanking the product and engineering teams."

These constraints help us produce results that provide the exact format that the user requires.

Tone

This is also a very important component while generating a prompt because it will provide us with the kind of tone that we need to use for a particular task. Consider the example, which is a continuation of the above prompt.

Ex_prompt4: "Use clear and concise language and write in a confident yet friendly tone."

This will ensure that the content generated sounds formal and respectful to the person (for example, the boss).

Thus, these are the various components that we must focus on while writing the prompt in order to get better results. Along these components, we also have a few other techniques that are considered advanced prompting techniques.

The objective of our paper is to:

- Discuss the importance of prompt engineering and its positive impact on healthcare.
- Discover prompting methodologies that help grab better results from LLMs.
- Showcase how prompt engineering can be integrated with AI for cancer, heart, and other healthcare treatments.
- Contribute knowledge about the benefits of this emerging technology to co-researchers and society.

Overall, this paper gives us a deeper understanding of prompt engineering, advanced prompting techniques, and the mission and vision of this technology. It also discusses how prompt engineering helps in disease diagnosis and treatment. In the end, we focus on the benefits and flaws of this technology along with the directions.

This paper is organized as follows: Sect. "Mission and Vision" focuses on the Mission and Vision of prompt engineering in healthcare. Section "Advanced Prompting Techniques" discusses Advanced prompting techniques. Section "Roles of Prompt Engineering in Healthcare" explains the opportunities, challenges, and issues of prompt engineering in healthcare. Section "Opportunities of Prompt Engineering in Healthcare" deals with how prompt engineering contributes to medical diagnosis and treatment. Section "Challenges of Prompt Engineering in Healthcare" suggests various advantages, disadvantages, and applications of prompt engineering. Section "Issues of Prompt Engineering in Healthcare" briefly describes ML algorithms that contribute to prompt engineering in healthcare. Section "Prompt Engineering in Cancer

Diagnosis and Treatment" concludes with future directions of prompt engineering in the healthcare domain.

Mission and Vision

Mission

Our mission is to explore the benefits of prompt engineering in the healthcare domain, focusing on exploring its impact on communication, patient outcomes, and better results. By diving deeper into the components and applications of prompts, we work to provide valuable insights that connect the gap between technology and healthcare. Our mission is to contribute knowledge that helps healthcare researchers and technology experts, where prompt engineering creates a space for more accurate, precise outcomes by giving patients a better experience in the healthcare domain [2–4].

Vision

Prompt engineering is rapidly advancing due to the enhancement of Artificial intelligence. This technology will be beneficial in many fields, including healthcare. Our research paper envisions improving human life by focusing on healthcare, including factors like quick and precise disease diagnosis, connectivity, and adaptability, and focusing on helping various healthcare researchers in their study. As shown in Fig. 2, various visions can be achieved by prompt engineering [4, 5].

Reducing the Diagnosis Time

Our vision is to reduce the diagnosis time, as the diagnosis is the crucial step in the treatment process. Advancing this will save time and increase the chances of patients' survival in a few cases. As the model is pre-trained with valid datasets,

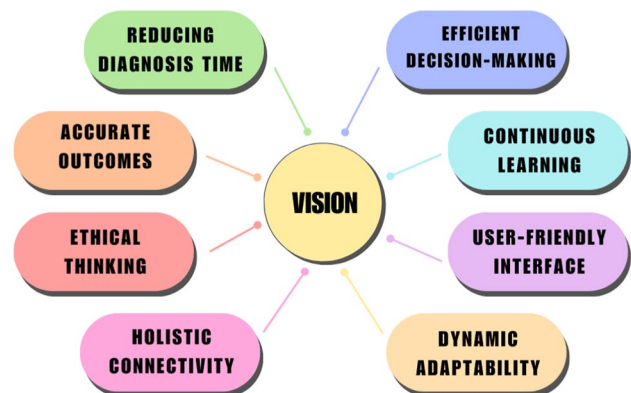


Fig. 2 Vision of prompt engineering in healthcare

we will compare the patient's reports with the model. Thus, the results are generated immediately, ensuring time and life-saving.

Obtaining more Precise and Accurate Outcomes

The goal is to enhance the accuracy of the results so the patients can confidently rely on the model. This can be achieved only by providing a large number of input datasets. Once the models are well-trained, achieving better outcomes through various prompting techniques becomes much more accessible. So, prompting helps us gain great insights from various patient data.

Efficient Decision-Making

In healthcare, the focus is not only on making decisions faster but also on ensuring the results are effective. By using prompt engineering tools and techniques, we empower healthcare teams to fasten their decision-making processes. This results in effective outcomes, thus ultimately improving patient care.

Continuous Learning and Evolution

Here, our focus is on creating solutions that can easily adjust to new challenges in healthcare. This ensures that the models are up to date and meet the latest requirements required by healthcare experts. This process involves learning data-driven insights from both current and past interactions.

User-Friendly Interface

A user-friendly interface will make the tools more accessible for healthcare experts. Once they are capable of operating these tools, the industry becomes more productive. Additionally, user-friendly interfaces will help the users with real-time guidance and support. This also ensures feedback and an interactive mechanism in the treatment process.

Ethical Thinking

Along with precision and accuracy, it is also very important to uphold ethical standards. This field involves dealing with a lot of personal data, such as patient confidentiality and personal and demographic information. We need to focus on the well-being of patients and respect their rights and dignity. We also need to ensure that the results generated are unambiguous and transparent.

Holistic Connectivity

In the context of prompt engineering, holistic connectivity refers to the creation of a network that connects patients with healthcare experts. This interconnection facilitates real-time communication and collaboration among doctors and patients, breaking barriers, promoting continuous care, and improving patient engagement.

Dynamic Adaptability

Here, our focus is on creating solutions that can easily adjust to new challenges in the healthcare sector. It is most important to make sure that the model being used is highly effective and works according to the latest requirements for the Domain. When the model fails to adapt to changing needs, it instead gives negative results.

The mission and vision of our research are to improve human life in healthcare. By achieving these, Domain makes human life much better and improves the quality of living by transforming the whole healthcare system from disease identification to better treatment [3, 6].

Advanced Prompting Techniques

The above-mentioned are the basic prompting component; as the field is evolving so fast, we have a few more advanced prompting techniques to make the results from LLMs more productive and highly efficient. The advanced techniques will be discussed in detail.

Zero-Shot Prompting

In Zero-Shot Prompting, the LLMs can respond to the users without prior training. No data will be given to the model, making it simple to use. However, the drawback is that we need to get the required outputs, so this is used for elementary tasks. It is always important to remember that the more data you provide, the more accurate the results will be. Let us consider the prompt below [1, 5, 7].

Ex_prompt:

Tell me whether the text is neutral, negative or positive.
“The new software had some great features, but I’m still trying to find how to use them.”

Sentiment:

Output: Neutral

Zero-Shot Prompting Usage in Medical Diagnosis Scenario (Breathing Problem)

In medical diagnosis scenarios, Zero-Shot Prompting can be useful for initial symptom detection in patients. For example,

when a patient shows symptoms like continuous cough or difficulty breathing a zero-shot prompting model can give an initial classification such as "breathing problem" [8, 9]. This method may be particularly useful in emergencies that require quick decisions to be made. However, one limitation is that it is not specific enough to work well without giving more context and examples.

One-Shot Prompting

In this technique, the model will be provided with a single example, which will help the model understand the context and make the results generated slightly better than zero-shot prompting. These are primarily used in real-time scenarios to use models like ChatGPT. Let us consider the example below; we have given an example prompt on how we expect the result. Thus, the model uses it as a format and generates the corresponding result [1, 5, 7].

Ex_prompt:

Translate the following English text to German:
 "Hello, how are you today?" - "Hallo, wie geht es Ihnen heute?"
 "What are you doing?"
 Output: "Was machst du?"

One-Shot Prompting Usage in Medical Diagnosis Scenario (Lung Cancer)

The One-Shot Prompting gives the model a single example, which helps in a basic understanding of the scenario. In medical diagnosis too this model can help initially by providing an instance. For instance, prompt such as "Identify the condition based on these symptoms: 'A 55-year-old male with a history of smoking presents with persistent cough and weight loss. The example diagnosis is lung cancer [10].' The model will then apply this feedback to new or similar cases. This method will improve the diagnosis process in healthcare by utilizing the model's results from the past and keep improving by using the various medical patterns, resulting in a much better approach when compared to zero-shot prompting.

Few-Shot Prompting

In Few-Shot Prompting, we will provide additional context. These models will be able to respond in a much-specified way as required. Here, we will provide more than one context/format to get concise and accurate results. The example prompt below will help you understand a few shot prompts. Initially, we provided the example data along with the format, and then we requested another example following a similar format [1, 5, 7].

Ex_prompt:

A 'glompit' is a rare plant which is found in the rainforests of Brazil. The example of a sentence which use the word glompit is: During our expedition, we stumbled upon a vibrant glompit hidden among the lush foliage.

To "snizzle" is the act of skillfully balancing objects on your nose. An example of a sentence that use the word snizzle is:

Output: At the talent show, she amazed everyone by managing to snizzle a stack of books without dropping a single one.

Few-Shot Prompting Usage in Medical Diagnosis Scenario (Hypothyroidism)

In the few-shot prompt model, comparisons can be made with zero-shot and one-shot methods. This is a technique that will help the model understand and improve its responses more precisely. Imagine a scenario of making a medical diagnosis where Few-Shot Prompting can be useful for differential diagnosis. For instance, consider the following prompt that contains several examples of symptom sets and their corresponding diagnoses:

Sample 1: "A middle-aged woman complains of fatigue and dry skin as well as weight gain leading to Hypothyroidism being diagnosed."

Sample 2: "An old man urinates often, feels very thirsty, and has lost weight unintentionally; this could be Diabetes Mellitus."

From these illustrations, whenever given a different set of symptoms like "A 50-year-old woman complains of palpitations, sweating after light activity, heat intolerance," it can easily predict or rather "Hyperthyroidism" [11, 12]. The method enables the model to make use of various examples so that it becomes more dependable.

Chain-of-Thought Prompting (CoT):

This advanced technique will make the model think and evaluate the problem accordingly. The evaluation is done in a sequence of intermediate steps. Consider the example below; the model needs to evaluate the problem in a particular order using complex reasoning strategies. This will break down the complex problem into more minor problems. This also uses the concept of Few-shot prompting. Then, it will understand the evaluation pattern and then produce the results [1, 5, 7].

Ex_prompt:

Prompt1: The sum of the prime numbers in the list is greater than 30: 2, 5, 11, 8, 6, 17, 3.

A: Adding all the prime numbers (2, 5, 11, 17, 3) gives 38.

The answer is True.

Prompt2: The sum of the prime numbers in this list is greater than 30: 7, 4, 9, 14, 6, 3.

A: Adding all the prime numbers (7, 3) gives 10.

The answer is False.

Prompt3: The sum of the prime numbers in this list is greater than 30: 23, 5, 13, 8, 6, 17?

A:

Output:

Adding all the prime numbers (23, 5, 13, 17, 3) gives 61.

The answer is True.

Chain-of-Thought Prompting (CoT) Usage in Medical Diagnosis Scenario (Myocardial Infarction)

This model breaks down the complex problem into simpler steps. In the case of a medical scenario, this method will help in evaluating multifactorial conditions. For example, a model tasked with diagnosing “chest pain, shortness of breath, and fatigue” in a patient would be steered by CoT prompting as follows:

The first thing for the model to do is to analyse the patient’s history and risk factors like age, way of life, and family history. Also, it considers different diagnoses like myocardial infarction [13], pulmonary embolism, or anxiety disorder. Finally moving on to identifying appropriate diagnostic tests such as ECG, D-dimer test, and chest X-ray.

The step-by-step analysis will help improve the accuracy of the process of diagnosis. When these symptoms are found with myocardial infarction, the model will say: “Based on the presence of chest pain combined with risks such as age and lifestyle so likely diagnosis is Myocardial Infarction. Recommend immediate ECG and cardiac enzyme tests.”

Self-Consistency

The self-consistency technique is prime for all the other prompting techniques. This is determined in order to enhance the chain-of-thought method. Using this technique, we can improve the model’s reasoning, arithmetic, and fundamental sense. Self-consistency will select ‘the better result from number of results that are generated.’ This further enhances the quality of the model. Making it an essential factor in all other prompting techniques [1, 5, 7].

Ex_Prompt:

Back in 2005, the total population of a town was 10,000. The population increased by 2% each year. What is the current population in 2023?

Output:

In 2005, the population was 10,000. With a 2% increase each year, the population grows by 200 each year ($10,000 * 0.02$). From 2005 to 2023, there are 18 years. So, the total increase in population is $18 * 200 = 3,600$. Adding this to the initial population, the current population in 2023 is $10,000 + 3,600 = 13,600$.

This example has solved the actual prompt and many sub-problems. They are making it easy to understand and making the generated results correct. The same problem is given multiple times to the model. All the results it has generated are stored, and the most common and consistent solution is returned as the result [1, 5, 7].

Self-Consistency Usage in Medical Diagnosis Scenario (Medical Dosages)

The Self-Consistency technique will make the model more reliable because it evaluates a number of responses and gives the best one among all the others. In medical scenarios, this model can be largely useful as it provides a lot of consistency and accuracy in responses. For example, let us consider a scenario where the model is asked to estimate the dosage of a medication to be given based on patient data [14]:

Ex_Prompt:

A 70-year-old patient with chronic kidney disease and hypertension needs a new dosage of medication X. The current dosage is 50 mg daily, but renal function has declined.

The corresponding outputs from the model include different suggested dosage such as 45 mg, 40 mg, and 42 mg. By applying Self-Consistency, the model compares these outputs and then selects those which are more consistent such as 42 mg thus becoming more confident about its recommendations. This method also helps reduce variation and improve the decision-making of the model, especially in critical medical situations that may have fatal consequences if they get the wrong medication dosages.

Generated Knowledge Prompting

This advanced technique focuses on generating information from LLM by using all the data that has been given as datasets(input) for building the model. Its workings are quite different from those of other techniques. Generated Knowledge Prompting is further classified into two types.

1. *Generate knowledge:* This involves generating factual data. In other words, this method will work on the facts/data on which it has already been trained.
2. *Using the generated knowledge:* This will use the knowledge it has gained from generating the results for

older prompts. This will be helpful when replying to multiple questions that belong to the same input data.

This prompting technique is considered the most difficult to implement as it requires more understanding of the model's training datasets [1, 5, 7].

Ex_prompt:

Question: Who wrote the play "Romeo and Juliet"?

LLM without GKP: I'm not sure about that.

LLM with GKP:

Step 1: Generate knowledge

Prompt: Create a list of famous literary works and their authors.

Output: "Romeo and Juliet" by William Shakespeare
"1984" by George Orwell

Step 2: Use generated knowledge to improve performance

Prompt: Who wrote the play "Romeo and Juliet"?

Output: William Shakespeare

Generated Knowledge Prompting Usage in Medical Diagnosis Scenario (Treatment Suggestions)

The Generated Knowledge Prompting (GKP) utilizes internal knowledge from the model to which it was trained for answering queries that are not known. GKP can be used in medical scenarios to provide further information on diseases, treatments, and medicines. For example, when a model is asked about what treatment should be given in a certain condition [15]:

Ex_prompt:

"Question: What are the treatment options for Stage II breast cancer?"

LLM without GKP:

"I am not sure what the treatment options are."

LLM with GKP:

Step 1: Generating the knowledge.

Prompt: "List out the common treatments for different stages of breast cancer."

Output: "Stage I: Surgery, radiation therapy. Stage II: Surgery, chemotherapy, hormone therapy."

Step 2: Using generated knowledge.

Prompt: "What are the treatment options for Stage II breast cancer?"

Output: "This means that stage two breast cancer is normally treated with a combination of surgical removal/ excision, chemotherapy, and hormonal therapies depending on specific tumor properties and patient factors."

This method ensures that the model provides not only accurate results but also utilizes its knowledge in the medical domain.

Tree of Thoughts (ToT)

It also improves the model's reasoning skills, just like the chain of thought prompting. How the model behaves differs from CoT because when we use ToT, the model goes much deeper than CoT and makes many comparisons. "The working of ToT is it finds out a solution for the first instance and then stores it in a tree in the form of thoughts; later, it finds out the result for the corresponding subproblem until all the subproblems are solved; once the result is found, it will generate the final efficient output from the tree it has generated [1, 5, 7].

Ex_prompt:

Scenario: Imagine three experts analysing a business expansion strategy.

Expert 1: Identify potential markets with high demand for the product or service. (//Based on the output from here, it utilizes the knowledge in sub sequent steps evaluation.)

Expert 2: Evaluate the regulatory environment and business policies in the selected markets to ensure compliance and minimize risks.

Expert 3: Conduct market research to understand local consumer preferences, competition, and pricing dynamics in the chosen markets.

These steps are evaluated step by step, after each step if the expert has provided a wrong information, then he will be removed. So here sequence of evaluation is done by considering knowledge from previous steps. Here is an image representing the working of ToT technique.

Tree of Thoughts (ToT) Prompting Usage in Medical Diagnosis Scenario (Cardiac & Respiratory Diseases)

The Tree of Thoughts (ToT) technique works just like the decision tree. The model organizes symptoms into a 'tree' of potential conditions and their accompanying signs. In this way, the model can explore different branches to identify the most suitable case that matches the required output.

For example, let us consider a patient with chest pain, shortness of breath, and a history of smoking. This ToT model can then take into account issues such as possible cardiac causes (angina, myocardial infarction), respiratory causes (COPD, pneumonia), or non-medical factors like anxiety [16].

The clinical data works separately in any of these ways removing unwanted branches and leaving only the desired pathway for diagnosis like coronary artery disease or pulmonary embolism. This way helps to get accurate results and assists in choosing appropriate treatment options.

Retrieval Augmented Generation (RAG)

It is a much-advanced technique that enhances the results generated and makes the LLMs more knowledgeable. This technique involves updating the data sets with the latest information and fine-tuning the model. Meta-AI researchers introduced this technique, which will combine the retrieval component with text-generating models. This takes input from various sources and merges them all by selecting the most repeated ones using a few existing techniques like self-consistency. As the model is fine-tuned, this results in generating efficient outcomes.

This technique is used in most popular LLMs like DALL-E 3, GPT-4(ChatGPT), and Mid Journey, generating the most consistent and up-to-date data [7].

Thus, these Advanced Prompting methods are crucial in prompt engineering technology. They are used according to scenarios to get accurate results from AI and LLM models.

Retrieval Augmented Generation (RAG) Prompting Usage in Medical Diagnosis Scenario (Metastatic Melanoma)

The Retrieval Augmented Generation (RAG) model in healthcare can be highly useful because it has capabilities to access the external database which can be useful to extract real-time information and generate valuable results. These models are capable of retrieving any information like medical documentation, clinical guidelines, EHRs, etc.

For example, let's assume a situation where a doctor wants to find out the most modern treatment procedures for a particular disease such as metastatic melanoma. To deal with this, we can use the RAG model and access data from some of the most current studies. Later on, it filters the information needed and gives answers. The capacity to access up-to-date information will enable healthcare providers to make early diagnoses and recommend appropriate therapy. Thus, improving the overall patient outcomes and making the healthcare domain more advanced and more reliable [17].

Roles of Prompt Engineering in Healthcare

Prompt Engineering plays a vital role in healthcare; due to its strong connection with Machine learning (ML) and Artificial intelligence (AI), it can perform huge tasks much quicker and efficiently. It has several roles that contribute to healthcare, from enhancing communication to focusing on patient satisfaction, from monitoring patients to international collaborations. We have now listed some pivotal roles that Prompt engineering will fulfil in healthcare [4, 6].

Integrating with Conventional Healthcare System

Integration involves combining conventional methods with the latest trends in healthcare to attain better results. When we integrate them, they work as one, making complex tasks simple. During the integration process, we provide many electronic health records (EHR), laboratory information, and various test results to improve the model's quality, which enhances the overall results of the model.

Personalized Patient Care

Personalized patient care is critical to treating individual patients with their medication style, as each patient's health condition varies from others. Here, prompt engineering shows its presence by suggesting health experts a unique medication for each patient based on his reports and health condition. This improves the chances of survival and recovery for the patient by minimizing the adverse effects.

Remote Monitoring of Patient

This helps patients and doctors from taking and giving treatment from any location. Using the intelligent wearable, the machine tracks the patient's health condition. It provides real-time updates to experts to monitor the patient's health even when not admitted to the hospital. This will be useful in a few cases as some patients do not like the hospital environment.

International Collaboration

Prompt engineering also facilitates extensive collaboration among several healthcare experts from different locations across the globe. Creating platforms where the experts can share the latest achievements and updates in the domain will be extremely helpful in advancing the whole healthcare

sector. This ensures a great foundation for knowledge sharing, and healthcare experts can enhance their skills.

Healthcare Analysis

Healthcare analysis also plays a crucial role, as the healthcare domain always requires enhancements. This can be achieved using prompting, where we can get insights from past patients' data and create valuable visualization data, which will help improve decision-making in the future. It also assists healthcare organizations in identifying trends and various patterns.

Enhanced Communication

Prompt engineering is crucial in exchanging information between patients and healthcare experts. Because it is the initial step in which the disease can be told to health experts to start the medication for the patient. So, enhancing communication is a crucial step in the whole process.

Patient Satisfaction

Prompt engineering also ensures patient satisfaction as it is designed to adapt and dynamically collect information like preferences and various feedback from several patients and use it to improve it. Prompt engineering always focuses on providing a great user experience, ensuring that it delivers quality and responsive healthcare services to the user. It also focuses on simple user interfaces to make it easy for the patients to access them, making them more straightforward and reliable.

These are the various roles of prompt engineering in the healthcare domain. These roles are designed to enhance human life in the healthcare domain. Bringing these into existence will truly redefine our healthcare sector [4, 6, 18].

Opportunities of Prompt Engineering in Healthcare

Using Prompt Engineering will provide numerous opportunities in many domains, including healthcare. This technology will ensure advancements in patient care, promise life safety, and reduce healthcare failures, which contributes to enhancing the whole healthcare domain. The opportunities include personalizing healthcare, remote patient monitoring, telemedicine, research, etc. [4, 19].

Personalized Healthcare

Prompt engineering can provide a unique medication style for a patient based on his health condition. Providing

patients' reports and current health status to the model along with suitable prompts will ensure personalized healthcare.

Ex_prompt: These are the reports of patient 'XYZ'. Study them carefully and remember that he is suffering from chronic obstructive pulmonary disease (COPD). Suggest a personalized healthcare medication for the patient to recover from the disease without any side effects.

Prompts like these will ensure that the patient is provided with accurate treatment without harming his life.

Remote Monitoring

Remote Monitoring will allow patients to receive treatment anywhere. Using the latest technologies, the patient's health condition is continuously monitored using smart devices. This will ensure that the patient is continuously monitored by the model. When an emergency occurs, the response is sent directly to the healthcare experts and also to the patient. Overall, this saves the patient time from being in the hospital, and he is also monitored continuously.

Improving Healthcare Knowledge

Prompt Engineering can also be used to build healthcare knowledge among people and many communities. One key reason patients get diseases is their lack of health awareness. So, we can use prompt engineering to deliver health safety messages, behavioural suggestions, medication, and lifestyle reminders for patients to prevent and cure diseases.

Telemedicine and Telehealth

Telemedicine and telehealth can boost the whole healthcare system. In general, teleservices refer to providing facilities in remote locations. Telemedicine means providing medical support, like sending medicine and other required medical apparatus to remote locations, and telehealth means providing various facilities like online appointments with doctors, medication, and patient care through virtual applications. In this process, the experts must identify the patient's health condition. To know the exact patient's condition, we can use prompting and AI technologies to provide real-time insights into a patient. Thus, teleservices can be provided to patients located remotely.

Research and Innovation

Prompt engineering is an excellent opportunity for researchers. As this era is an evolution of artificial intelligence and machine learning technologies, there is much scope for research in this field. The researchers can focus on subdomains like prompt engineering, which have a crucial impact on AI and ML. It is essential to integrate

prompt engineering and AI in every domain, along with healthcare. This research helps innovate new things that can improve human existence.

Quality Improvement

The other opportunity for prompt engineering is to improve the quality of existing models. This can be done using continuous evaluation through feedback mechanisms and fine-tuning the model. Along these lines, it is always essential to create something new in order to meet the latest domain requirements. As the healthcare domain keeps updating very fast, focusing on new innovations in this field is essential. So, prompt engineering can be used to summarize large datasets. Thus, domain experts can work directly on the development part rather than focus on data clustering.

These are the various opportunities that can be grabbed by using prompt engineering in health care. It is most important to use available resources to maximize the benefits [4, 6, 19].

Challenges of Prompt Engineering in Healthcare

Although prompt engineering and AI are the most popular technologies at the moment, some challenges still have to be addressed to make them more reliable. Challenges of prompt engineering include privacy concerns, organizational barriers, integration issues, and many more. Now, we have listed out the challenges of prompt engineering in healthcare [20, 21].

Data Quality

The model depends on data for quality results or accurate predictions. If the model is trained with correct data, then the results are accurate and lead to correct information, which can be lethal in healthcare. So, training the model with the latest data and correct information is a huge challenge for healthcare experts. These models become better and better only with fine-tuning techniques. Fine-tuning involves training the existing model with more quality information. This is the only way in which we can overcome this challenge of data quality.

Privacy and Security Concerns

The prompt systems generally give patients records, reports, and personal information as input to discover any disease presence. This process of giving patient samples is crucial and risky. The patient's data can be at risk when a data breach occurs. This can be very dangerous for the

patient's life and affect the trust in the healthcare domain. So, privacy and security concerns are other challenges in this technology. It is essential to encrypt the data using security algorithms that will make the prompting system more dependable.

Acceptance by the User

Both patients and healthcare experts are habituated to the traditional healthcare system. Adapting to the latest prompting system techniques might be challenging for experts. Using the latest methodologies for treatment might sometimes cause patients to feel unsafe. To overcome these challenges, patients and domain experts must be made aware of the benefits of the latest techniques. So, this can be very challenging. This can be overcome only by promoting various benefits of prompting in healthcare.

Training Healthcare practitioners

The other challenge includes training healthcare experts on using the latest equipment (advanced medical tools). Most physicians, nurses, and healthcare professionals must learn to use the latest tools. So, training experts with these advanced machines involves a lot of cost and time. This also acts as a significant challenge to integrate prompting in healthcare. However, the results when the experts are trained will be very high. So, Healthcare organizations should conduct mass training on using tools to overcome costs and train medical teams all at once.

Integration of Technology

Integrating prompt engineering in healthcare can be complicated and challenging. Most medical equipment should be replaced with the latest and advanced tools, which are expensive and difficult for medical centers. This is a more significant challenge than any other technology. To overcome this, medical teams should start prioritizing the better results produced by prompting over cost and expenses. These should be treated as investments in infrastructure upgrades in the medical centers.

Cultural and Organizational Barriers

Along with integrating technology, cultural and organizational barriers can be very challenging in enhancing technology. Because the majority of top organizations run hierarchically, and acceptance should come from higher officials. This can take much time, and in a few cases, it causes cultural issues like a lack of resources, skilled professionals, and funds. The only way to overcome this challenge is by making organizations aware of the

benefits that can be obtained by using the latest healthcare methodologies by integrating prompt engineering techniques.

These are the various challenges that are faced in prompt engineering. To overcome these, the healthcare sectors and people must be aware of various benefits that can be drawn from the prompting technology when integrated with the healthcare domain [20, 21].

Issues of Prompt Engineering in Healthcare

Considering various factors of prompt engineering in healthcare, this technology has pros and cons. Although this technology significantly impacts health, a few aspects have to be considered as they stand as significant issues in this field. Some of the issues of prompting in healthcare include unfair responses and unpredictability, misuse of data, loss of decision-making, integration issues, and many more. As shown in Fig. 3, these are some significant issues related to prompt engineering in healthcare [6, 20, 21].

Unfairness

Occasionally, prompting systems can generate false information, which can be caused by training the model with incorrect information or poor algorithm performance. When healthcare experts rely too much on these models, patient life can become challenging, which is very dangerous. So, this is one of the most problematic issues in prompt engineering. To overcome these, the models have to be fine-tuned multiple times to improve their results and also tested on various samples before being released for use.

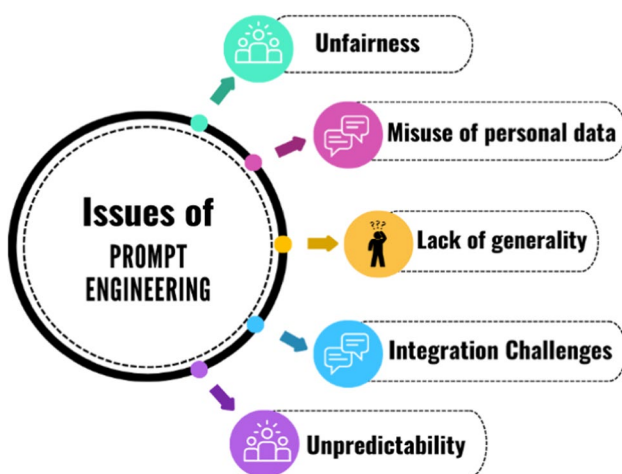


Fig. 3 Issues of prompt engineering

Misuse of Personal Data

The other issue of prompting in healthcare is misuse of personal data. As this field involves providing patients' personal information to the machine, it is equally important to protect it. In case of a data breach, the patient's personal information might be stolen and used for illegal activities. Experts and LLM developers must prevent these issues to make the technology safer. Overcoming this issue will make the prompting technology in healthcare more reliable.

Negative Impact on Health

There are chances of negative impacts on health because of over-dependence on AI and prompting. These models are trained to send alerts when they find any irregularity in a patient's health. They might occasionally send alerts in false scenarios, resulting in the patient getting confused and traumatized. These results in negative impacts on patients' health, which disturbs the patient's mental health. These issues should be overcome to make technology safer for patients.

Loss of Human Decision-Making

Relying on technology too much will slowly impact human's IQ and thinking capabilities. Usually, doctors perform various tests to study those reports and decide the treatment for the patient. However, if prompting technology is entirely implemented in healthcare, it can replace everything from performing tests and studying them to suggesting treatment to the patient. This reduces the burden on healthcare experts, but in the long term, this will make doctors over-rely on machines, resulting in a loss of critical thinking capabilities and decision-making skills.

Unpredictability

The machines trained might sometimes produce uncertain data. There are a few reasons behind this issue; one is a lack of quality data, and the other is a lack of algorithm functionality. Thus, these become highly unpredictable in a few cases. These issues must be addressed in the testing stages of the model, and then the model should be released for real-time usage in the medical sector. In the future, the models can improve and overcome these issues because of research and innovation taking place in the world right now.

Integration Challenges

Another major issue in healthcare is integrating the latest models in the existing healthcare domain. This becomes very difficult and expensive as most healthcare

organizations want to go with existing techniques instead of adding new machines, which can overprice. The scans and reports produced with existing machines might not be compatible with new ones. The other integration issues include patient acceptance of new models or machines for treatment.

Lack of Generality

In prompt engineering, lack of generality means the models must work better for all cases or all patients. This happens when the models are trained for a specific disease or situation. For example, a model designed for heart care might not work well for other diseases. To overcome this, the model should be trained for all the possibilities that are possible during the testing process. This is the only way in which we can overcome the issue of lack of generality.

Thus, these are the most commonly addressed issues of prompt engineering in health care. These must be overcome to make the prompting-based models more effective and reliable in any domain, including healthcare [6, 20–22].

Prompt Engineering in Cancer Diagnosis and Treatment

The process of cancer detection and treatment involves several sequential steps aimed at early identification and personalized management of the disease.

Diagnosis

The first step in cancer treatment involves diagnosing cancer in the patient. The conventional methods used in this process are biopsies, image scans, mammogram screening, and blood and bone marrow tests. These methods have been used in cancer detection over time. These conventional methods cause much pain for patients during the tests, especially biopsies, which involve collecting a sample of cell tissues from the cancer suspecting area in the body. The rapid evolution of technology has brought new techniques to overcome the drawbacks of conventional methods. So, these latest methods will ensure the diagnosis is fast, simple, accurate, and pain-free.

Due to the evolution of new technologies like Prompt Engineering and Artificial Intelligence, it is possible to detect and predict in the future whether there is a chance for the patient to get cancer. This can be achieved by training the models (LLMs) with data that can accurately predict the presence of cancer in the human body [23–25].

Using Prompt Engineering in Cancer Diagnosis

The diagnosis process with prompt engineering has changed; here, the scans are taken from cancer-suspecting areas in the patient. These are then given as input to the model, designed for cancer detection, and work with prompts from healthcare experts and test reports. Then, the model starts comparing the reports with the datasets on which it is trained. When it finds any patterns related to cancer, it indicates and suggests to the oncologist the presence of a tumor; if there are no traces found, it checks for any chances of future cancer occurrence in a patient. This can only be possible if the experts give exact prompts that the developed model can understand. Thus, it ensures that cancer diagnosis is more reliable and precise with prompt engineering [23, 24].

Staging and Planning Treatment

The next major step that involves cancer treatment is staging. This step involves finding out the stage in which the patient with cancer is currently suffering. Usually, the cancer stages are from stage zero (0) to stage four (4). The intensity of cancer increases gradually from stage 0 to stage 4. In stage zero, the traces of cancer are hard to identify in the body. With the latest technologies like Artificial Intelligence and Prompt Engineering, Finding the presence of cancer can be diagnosed and staged in very early stages, which will be life-saving for many patients.

After identifying the stage of cancer using prompting, healthcare experts will start moving to the next step, planning treatment for each patient. Earlier, the treatment process used to be the same for every patient as doctors could not identify at which stage of the cancer the patient was suffering. However, with this technology, doctors can determine which stage the patient is in and plan better treatment accordingly. This increases the chances for the patient to survive and reduces side effects caused during the treatment [23, 24].

Treatment for Cancer

After planning the treatment for the patient, the experts start the treatment to cure the cancer. There are many treatments that have been used for a long time. The most used cancer treatments are Chemotherapy, Radiation Therapy, Hormone Therapy, etc. These methods are very hard to perform on patients, and there will be many side effects due to these treatments.

Chemotherapy: This treatment involves using drugs and steroids to reduce the growth of cancer cells in the body. These steroids are very harmful when used for a longer time. Thus, this treatment can cause many side effects to the patient, although it slows the spreading rate of cancer.

Radiation Therapy: In radiation therapy, very high doses of radiation are targeted at cancer tumors to destroy them. This therapy is hazardous as it might have side effects like skin irritation and hair loss in the treated area.

Hormone Therapy: Although hormone therapy is not used that often, it is an effective treatment for hormone-sensitive cancers such as breast cancer and prostate cancer. This therapy will involve blocking hormones that are helping the cancer to grow. Using this treatment can sometimes cause side effects like weight gain, osteoporosis, fatigue, etc.

Using Prompt Engineering in Cancer Treatment

Using prompt engineering will ensure the proper treatment is being given to the patient based on his current health condition. It also suggests how to perform the procedure, what precautions to take, and the amount of drug dosages to be given during the medication process. Earlier detection and giving the proper treatment would be difficult for healthcare experts as the reports were not accurate. However, with this technology, things have become quite simple in healthcare. From diagnosing to choosing the proper treatment, from helping experts in the treatment to monitoring patient health conditions, prompt engineering, and artificial intelligence have made significant changes in the healthcare domain. All the tasks can be done quickly by giving the correct prompts to the LLMs [23, 24].

Monitoring and Supporting Care

Post-completion of treatment, regular monitoring of patients is very crucial in cancer as there are chances of the reappearance of tumors even after surgery is performed. To overcome reappearance, healthcare experts perform monitoring to detect in case of cancer appearing again in the body. In this process, the patient must come to hospitals and cancer clinic centers often, daily or weekly. This process can be reduced to some extent using prompting and artificial intelligence [23, 24].

Using Prompt Engineering in Patient Monitoring and Care

We can use prompt engineering to integrate artificial intelligence with intelligent wearables to track patient health conditions. This can reduce unnecessary visits of patients to hospitals. These devices can accurately predict real-time data from a human and send information to domain experts to guide patients in an emergency. Along with monitoring patients, it is equally essential to provide necessary care to patients. This can also be achieved using prompting; these models can be designed in a way that they are capable of creating proper supportive care for each patient. So, based

on the patient's current health status, they will have their medication and care [23, 24].

Prompt Engineering in 'Breast Cancer Screening'

Breast cancer treatment starts with diagnosis, which includes performing imaging tests like mammograms, ultrasound, biopsy, and MRI scans. These reports are then given as input to the LLM, along with prompts to study these reports. These models identify many patterns and find out any presence of cancer tumors in the breast. Once the model examines mammograms, it generates output based on the severity of the infection. Then, the task becomes easy for healthcare experts, who can initiate the treatment process [23, 24].

Studies are showing that in the future, the models will be able to perform the treatment directly without any doctor intervention. But for now, the models are developed to guide doctors in performing treatment and taking precautions to avoid any side effects from the treatment.

The above Figs. 4, 5, 6 shows the reports of breast cancer presence. These reports can be used to train and test the model and achieve better results in healthcare. So, to use them in cancer screening, these models can be fed into the machine along with prompts that request the model to test whether there is any presence of carcinoma. The model will detect the cancer if patterns are identified in given samples.

Currently, breast cancer treatment is done in two stages; the first one is surgery, which is done using techniques like Lumpectomy and Mastectomy. Post completion of this surgery, there are few cases of cancer recurrence; to

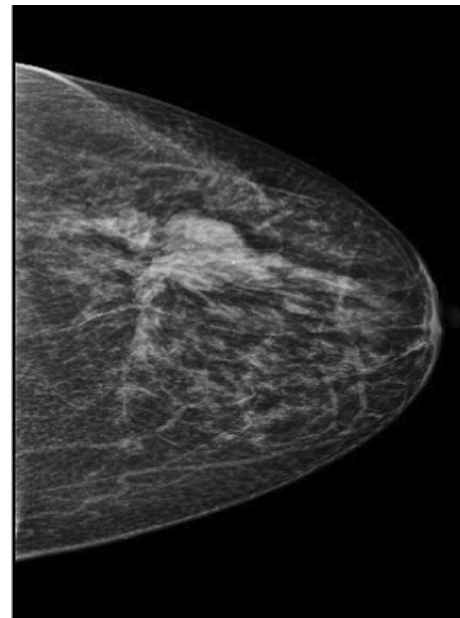


Fig. 4 Presence of spiculated left breast nodule (mass on left breast)

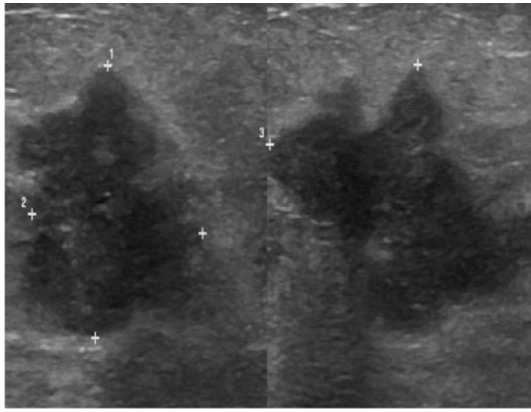


Fig. 5 Presence of external upper quadrant irregular, heterogenous, indistinct margin left breast mass

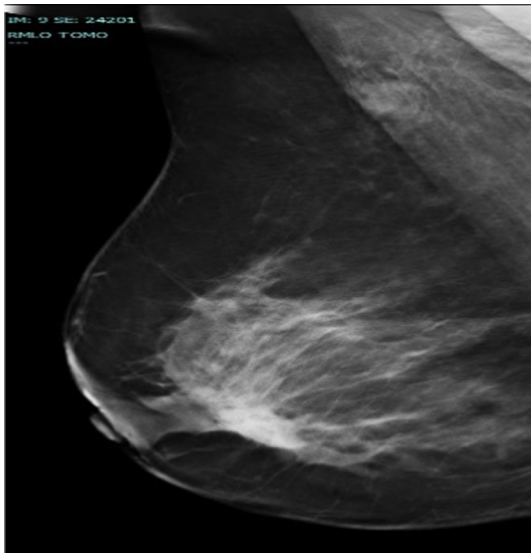


Fig. 6 An irregular, hyperdense, retro areolar mass with spiculated margins is seen in the lower inner quadrant of the right breast

tackle those situations, some patients might have to undergo subsequent therapy. The second one is Adjuvant therapy, which is used to reduce the chance of cancer recurrence. This includes techniques like radiation therapy, Chemotherapy, Hormone therapy, etc. However, these techniques will have serious side effects, as mentioned above. In the future, with much more advancement in AI and Prompting techniques, there are chances of reducing these effects [23, 24].

Prompt Engineering in ‘Lung Cancer Screening’

One of the other most dangerous cancers in the world is Lung cancer. The hardest part of lung cancer treatment is diagnosing its presence, making it very difficult to cure in the early stages. It is primarily diagnosed in the middle and

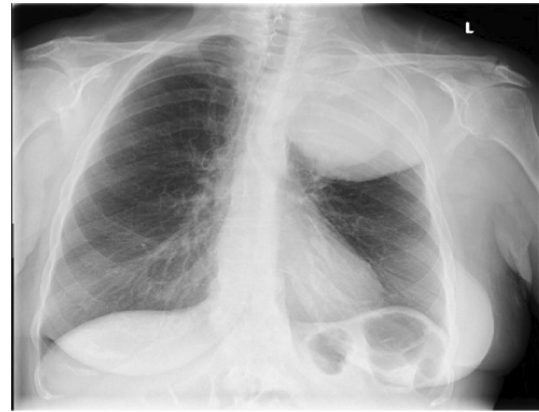


Fig. 7 Huge left upper lobe mass with destruction of the left third rib



Fig. 8 Bilateral calcified and non-calcified pleural plaques

late stages, causing death in most patients. Similar to most cancers, Lung cancer treatment also starts with diagnosis. If diagnosed, the medical team will start treatment based on the cancer stage. It includes techniques like Lobectomy and pneumonectomy. These techniques involve either removing a lobe from the lung containing a tumor (Lobectomy) or complete removal of the lung (pneumonectomy) [23, 26].

The above Figs. 7 and 8 are samples of lung cancer screening. These large-scale images are used to train the model. Once the model is trained with correct data, it can accurately predict the possibility of cancer presence and future occurrences. After cancer detection, the domain experts will start the treatment process.

These treatments involve using high radiation and powerful steroids, which cause long-term consequences to the patient. Prompt engineering and artificial intelligence help a lot in overcoming these drawbacks of conventional cancer treatment techniques as they are very accurate and precise in treating cancer tumours. One such model being tested is Sybil, which MIT scientists are developing. In

reports, it is stated that it works with an accuracy of close to 94% in tests. This model is designed exclusively for lung cancer detection, but in the future, it can be expanded to include many other cancer detection and treatment methods [23, 26].

Thus, this is how prompt engineering can help the healthcare domain in cancer treatment by integrating with artificial intelligence to grab better precise, accurate, and reliable results, ensuring patients' safety and reducing side effects. Prompting and LLM's will help make cancer treatment much more accessible and safer for humans [23–26].

Prompt Engineering in Heart Disease Diagnosis and Treatment

Heart diseases are one of the most common among all diseases throughout the world. These are very dangerous for the patients, as they are usually caused due to blockage of coronary arteries in heart vessels. Heart diseases are highly unpredictable because they do not show any symptoms in most cases. They are mostly found after getting a heart stroke for patients [27, 28].

In this section, we will explore early heart disease detection and screening and how prompt engineering can help medical experts in cardiology (Figs. 9, 10, 11).

Steps Involved in Heart Disease Detection and Treatment

Diagnosis

The first step in heart disease detection is diagnosing the presence; this is done through various methods, including ECG/EKG, Echocardiograms, Stress tests (TMT), Angiogram, MRI, etc. These are the conventional methods that are being used for diagnosing heart diseases. These methods can be performed based on the severity of the patient's condition. Usually, the diagnosis starts with an ECG/EKG, and then, based on the results, the doctors proceed for further tests.

Over the years, these techniques have improved their results, but in a few cases, when reports show mild abnormalities, doctors might fail to identify such mild differences in reports [27–30].

Prompt Engineering in Heart Disease Diagnosis

As mentioned above, in a few cases, it is hard to identify the presence of abnormalities. Then, the trained models can be given prompts and reports to spot the irregularities in reports with great accuracy and few errors. As models are trained with extensive data sets, the models can even identify very



Fig. 9 Extensive bilateral pulmonary emboli with secondary signs of acute right heart strain



Fig. 10 Hypoplastic proximal pulmonary artery with retrograde blood flow seen on colour Doppler from the ductus arteriosus is demonstrated

small presences of blockages or abnormal heart rhythms in the heart. This makes the diagnosis faster and more accurate.

Latest Development in Heart Disease Diagnosis

A new technology that can detect heart diseases just by scanning our eyes is under trial. It can detect diseases that are present now or that might occur in the future. This tool or model scans the human eyes and makes accurate predictions. This tool is still under trial, i.e., 'CLAiR'; it just senses our eyes through a retinal lens and can accurately diagnose patients. Many such tools are being developed worldwide, but the primary skill required is prompt engineering to handle any model [27–30].

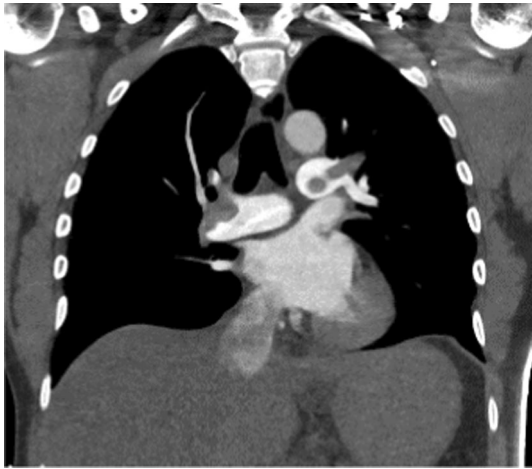


Fig. 11 Large saddle emboli extending into the lobar pulmonary arteries bilaterally. Right heart strain is present, with enlargement of the RV and bulging of the interventricular septum

Lifestyle Changes

Once a disease is diagnosed, the next step is performing immediate surgery or suggesting lifestyle changes for the patient. These changes are suggested when the intensity of the disease is low and can be cured with minor changes in his habits. These changes usually include changing dietary habits, such as reducing fat consumption and eating less carbohydrate food. Moreover, regular exercises like walking and other cardio exercises should be included. Healthcare experts make these suggestions [27–30].

Prompt Engineering in Suggesting Lifestyle Changes

Although doctors suggest them, many patients fail to follow them, which might threaten their lives again. So, we can use trained AI models that can successfully send you various alerts and suggestions to be followed. They can also track our progress when integrated into intelligent wearables. To perform any of these tasks, models must be given precise prompts to personalize results based on the patient's condition. We can use advanced prompting techniques, mentioned in our paper's 'advanced prompting techniques' section, to make models respond accurately and precisely.

Medications

When the intensity of the disease is too high and cannot be cured with lifestyle changes, doctors go with medication practices, which involve using cholesterol-lowering drugs (statins), blood pressure medicines, antiplatelet agents, and heart rhythm abnormalities controllers. This ensures that the patient's health situation comes under control. However,

there might be better practices than the medication phase when the severity of the disease is too high. Even in the medication phase, prompting models can effectively suggest to health experts about the dosage of use, side effects, etc. Prompt models can also suggest whether the medication style is suitable based on observing reports and scans. Thus, doctors can save patients' lives and time by treating them properly [27–30].

Interventional Procedures

When the intensity of the patient's condition is critical, domain experts suggest that the patient undergo an interventional procedure (surgery). There are many interventional procedures, but the most widely used are Angioplasty (PCI), Valvuloplasty, pacemaker implantation, Bypass (CABG), etc. These are the most commonly used interventional procedures [27–30].

Prompt Engineering in Interventional Procedures

Developed models can be extremely helpful to guide doctors while performing surgery. These models are very accurate and can react according to the situation. So, an AI-based system can give real-time instructions to doctors, which reduces errors in the surgery procedure. They can also give feedback, which helps doctors assess themselves and make improvements.

Monitoring

Although intervention(surgery) has been completed, there is a chance that a patient will be affected by heart disease again in the future. To ensure this, doctors usually suggest that patients follow a strict diet, have a healthy lifestyle, and have regular checkups. Using intelligent wearables, we can use prompting techniques to monitor the patient 24/7. They collect and monitor patient data continuously. In an emergency, they send alerts and inform healthcare experts about the situation [27–30].

Thus, prompt engineering helps in cardiology, making the whole process safer and extremely accurate. With continuous research and development in this field, not only heart care but also the whole healthcare system becomes much more reliable and safer.

Advantages and Disadvantages

Advantages

In today's world, technology is evolving rapidly. Thus, integrating the principles of prompting into healthcare results

in many advantages. Here, we have listed some of those exceptional advantages of integrating prompt engineering in the healthcare domain. Advantages include enhancing efficiency, cost and time saving, drug discovery, etc. We will discuss the advantages of prompt engineering in healthcare, as shown in Fig. 12 [6, 18, 20, 29, 30].

Enhanced Efficiency

The concept of enhanced efficiency is a crucial advantage in prompt engineering in healthcare. It reduces unnecessary delays in the disease detection process, making it very helpful in critical healthcare situations. Using prompting techniques will also ensure accuracy and precision in the test results. It also reduces the chances of fatality by diagnosing the diseases early and suggesting better treatment according to the patient's health status and records. Automating most tasks allows the experts to focus on high-level tasks and more research on healthcare.

Improved Patient Outcomes

Patient outcomes will improve when prompt engineering is used in healthcare. It identifies patterns, trends, and other risk factors in patients. Patients can be monitored from a remote location using prompt engineering. This will ensure the patient can live comfortably, thus gradually improving outcomes. It also sends timely notifications to patients about important updates and their health conditions. This ensures that the patient's chances of curing a disease increase significantly. This results in patient satisfaction and enhanced quality of life.

Decision Support

Prompting provides valuable healthcare information, which can be very helpful in making decisions from the machines. Usually, the machines are developed using large amounts of data to make accurate predictions. Thus, the more data we train with, the better decisions become. Along with

the data, algorithms also play a crucial role in making the predictions more precise. Thus, to make the algorithms more dependable, they must be trained on the latest data, ensuring proper decision-making in results.

Cost and Time Savings

Prompt engineering helps in saving cost and time as it can perform various tasks simultaneously. This will automate tasks like scheduling appointments, assisting patients with basic requirements, etc. By automating small tasks, it will ensure that there is less to no burden on healthcare experts. This will avoid over-dependency on health experts, thus making it cost-saving for the healthcare industry. Overall, prompt engineering can reduce the intervention of experts by replacing them and ensuring cost and time savings.

Drug Discovery and Development

This technology can boost the discovery of drugs and various developments in healthcare. It helps healthcare researchers by contributing the latest trends and information, which can lead the experts to focus more on clinical tests and drug discovery. Usually, experts spend much time collecting data samples for research, but with prompt engineering, data can be easily summarized. Thus, it has accelerated the pace of discovering innovations in the healthcare domain.

Data-Driven Insights

Prompt engineering is known for producing data-driven insights. It provides insights by collecting the patient's data, which includes test results, lab reports, images, X-rays, etc. These are compared with existing datasets, and the latest predictions are made on them, which will help in studying the area of concern in healthcare. This ability to adapt dynamically will help in obtaining more insights from data. This information can also make future predictions and act as life-saving in the healthcare industry.

Thus, these are various advantages of prompt engineering in healthcare, which ensures improving healthcare, making it more reliable and safer for patients. Implementing these in the current healthcare system will ensure that healthcare becomes more beneficial and safer [18, 29, 30].

Disadvantages

While prompt engineering in healthcare offers several benefits, it also has some drawbacks that must be considered for safety and should be overcome to make technology more reliable and safer. Here, we will list some of the disadvantages that stand as a barrier to technology. They include security concerns and disparities, from ethical issues to

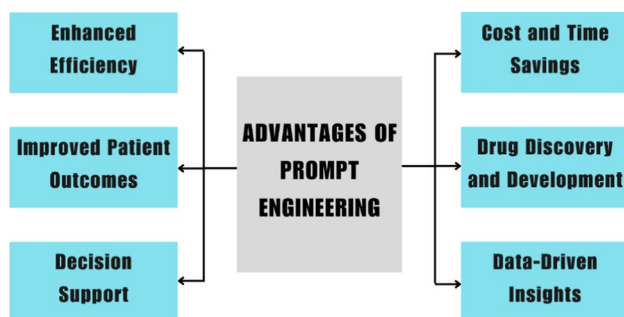


Fig. 12 Advantages of prompt engineering in healthcare

over-dependency on technology, as shown in Fig. 13 [12, 18, 20, 22, 31, 32].

Overwhelming Alerts

Using prompt engineering, we can also send alerts to patients related to important events and tasks. However, sometimes, when these alerts are not appropriately controlled, they become overwhelming. There are many disadvantages because these unnecessary alerts make patients worried about their health condition, and when they receive continuous alerts, they might also lose trust in them. It is essential to ensure that the model generates the alerts only during an emergency.

Privacy Concerns

Privacy concerns are a crucial drawback in a few cases of prompt engineering. The healthcare field requires storing a lot of patient data, including medical data, personal information, test results, etc. So, it is essential to store the data securely. If a data breach occurs and a patient's data is leaked, it will have severe consequences for the patient's life as well as the reputation of the healthcare industry. To overcome these, the prompting should be done more precisely. This reduces most of the privacy concerns issues.

Overdependency on Technology

The other drawback is the overdependency of technology in healthcare. When we over depend on technology, healthcare experts lose their ability to work independently. This has a huge impact on the healthcare perspective in the long run. To avoid these, experts have to test themselves regularly instead of completely relying on machines and technology.

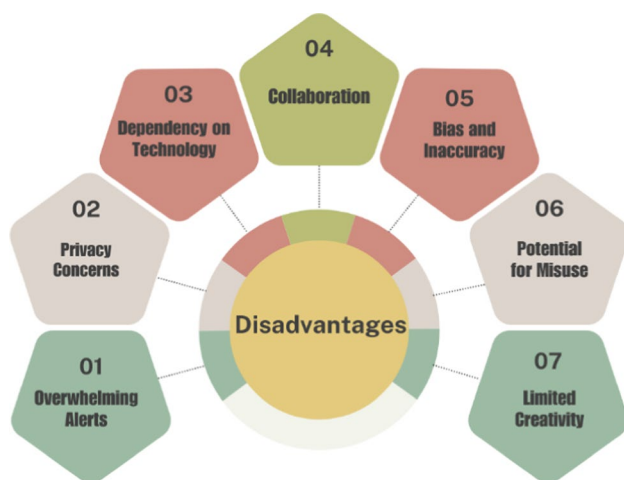


Fig. 13 Disadvantages of prompt engineering in healthcare

Collaboration

Collaboration has advantages and disadvantages, as it will be beneficial to share information among the healthcare experts in this domain and enhance the knowledge. However, there are also various barriers, such as language, methodologies being used, and culture among various healthcare experts. As each expert can have different strategies for solving similar problems, collaboration must be done to improve the knowledge among the healthcare experts instead of acting as a barrier.

Bias and Inaccuracy

Although the models are trained with vast datasets, there is still a high chance that the models will produce biased and inaccurate information. Healthcare experts need to verify the test results carefully after being generated. It is also essential to focus on the types of prompts given to the LLM. It is also essential to train the models according to the latest data to get more accurate results and less biased information from the LLMs when patients are provided with inaccurate information that also affects their trust in the healthcare system.

Potential for Misuse

With prompt engineering in healthcare, there is a chance of misusing the prompt system both intentionally and unintentionally by the people. These are built using a vast amount of data and various algorithms. These can work on patient data, including test results and medication information. So, there is a chance of misusing this data for commercial purposes. Consider the pharma industries using the data and suggesting unnecessary treatment to the patient will have dangerous impacts on the patient's life. As we all know, 'with great power comes great responsibility.' It is essential to ensure that data is correctly used [18, 31, 32].

Thus, these are the various disadvantages of prompt engineering in healthcare. Overcoming these aspects will ensure this new technology is more reliable and dependable for healthcare experts. It is very important to remember that technology should always be handled with proper care, as it always comes with both advantages and disadvantages.

Applications of Prompt Engineering in Healthcare

Prompt engineering has shown a significant presence in many domains, including healthcare. This technology in healthcare can completely revolutionize the healthcare domain. Integrating the prompting into existing systems will significantly impact human life. There are a lot of

applications of prompt engineering in healthcare that are life-saving for many patients. Its applications include respiratory disease management, cardiovascular risk assessments, maternal health, diabetes monitoring, and many more. These are some of the most popular applications of Prompt engineering in healthcare [6, 18].

Cardiovascular Risk Assessment

Prompt engineering can be extremely helpful in cardiovascular disease assessment. Cardiovascular refers to heart disease. The way how prompt engineering helps in cardiovascular diseases has been clearly explained in prompt engineering in the cancer diagnosis and screening section of our paper. They play a crucial role in determining the risk factors, as the whole treatment depends on risk factors for cardiovascular diseases. They are also accommodating in predicting the chances of heart attacks, strokes, and peripheral artery disease in the future.

Diabetes Management

The other popular application of prompting in healthcare is diabetic management. Prompt engineering can help patients monitor blood glucose levels, medication details, dietary plans, physical activity, and regular medical check-ups. When integrated with smart wearables, prompt engineering can also track blood pressure and sugar levels in real time. Then, the data is shared with the medical expert's team, who can assist with great medication for the patients.

Rheumatologic Disease Management

Prompt engineering can benefit rheumatologic disease management. It can help identify various diseases, including rheumatoid arthritis, lupus, and osteoarthritis. These prompt systems can help patients personalize their medication and read X-ray reports. When trained with large data, these systems are very accurate and precise. They also provide many essential lifestyle recommendations during the rest period of these diseases.

Respiratory Disease Management

Respiratory diseases such as asthma, chronic obstructive pulmonary disease (COPD), cystic fibrosis, and many others are a massive challenge for the patient's life. These are hard to detect during the early stages, as most lung diseases show common cold symptoms at the beginning, but as days progress, they get very severe. Prompting tools, when integrated with AI, can find out these diseases very early and provide accurate reports of lung samples, which are life-saving in many patients.

Cancer Detection and Screening

As seen in the paper, cancer detection becomes much safer and much easier in the future with the integration of prompting and AI. These can also predict the future occurrence of cancer in humans, which is truly a lifesaver for patients, and the task becomes more accessible for doctors. Much research is already being done on implementing this technology in healthcare. One such tool that is under trial is 'Sybil.' This tool can predict the likelihood of lung cancer occurrence in the future with an accuracy of 94 percent. So, patients can change their lifestyle to avoid the cancer.

Maternal and Child Health Monitoring

This technology does not end up identifying diseases but can also help monitor maternal and child health during pregnancy. We can easily monitor the baby's condition by integrating Prompting with AI. Along these, LLM's can be trained to schedule appointments, lifestyle suggestions, care to be taken, dietary plans, and many more. This can improve the life of the baby and the mother. When given good prompts and exact data, these models can perform the whole surgery independently, with health experts giving prompts to the machine. Compared to humans, when trained well, these LLMs are accurate and can have a smaller margin of error.

Infectious Disease Surveillance

Prompt engineering can be very helpful in tracking infectious diseases. These models can effectively make predictions using real-time data from various sources, including lab reports, disease outbreaks (COVID-19), blood samples, etc. Based on this data, the models immediately bring it to the attention of healthcare experts. So, healthcare experts can take quick actions like quarantining, mass vaccination rollouts, etc. This will finally protect the patient's life and the public's health.

Prompt engineering in healthcare can be helpful in many other healthcare applications as it is very accurate, precise, and powerful. Once these models are used, the whole healthcare domain will be redefined [18, 33].

ML Algorithms Used in Integrating Prompting Models

Machine learning algorithms play a crucial role in healthcare as they help perform various statistical operations on large amounts of data and provide us with great insights. Various machine learning algorithms are used in healthcare, including supervised, unsupervised, and reinforcement

algorithms. These combined help healthcare experts make early-disease diagnoses and improve patient healthcare. Here are the most widely used ML algorithms in healthcare [34–36].

Support Vector Machine (SVM)

SVM is one of the most widely used supervised learning algorithms capable of performing classification and regression tasks in the healthcare domain. These techniques are essential in minimizing errors while generating various outcomes. In healthcare, SVM is used primarily for disease diagnoses, predicting patient outcomes, and medical image classification. This is also known as the most widely used ML algorithm.

Neural Network

These are also artificial neural networks (ANN). They are another type of supervised learning algorithm that is being largely used in many fields, including healthcare. They work similarly to the human brain and can perform various thinking functionalities as a human does. By using techniques like forward and backward propagation, they learn to interrelate data and produce accurate outcomes. They are mainly used in healthcare for drug discovery and medical report analysis.

Logistic Regression

Logistic Regression is also a supervised learning algorithm used in healthcare to make accurate predictions. It uses a function called the logistic function to make these predictions and generate valuable and precise results. This logistic function continuously evaluates and keeps updating better possibilities. In healthcare, it is used to predict patient health results, disease classification, etc.

Decision Tree

Decision trees are another type of supervised learning algorithm used for complex analysis. They use the concept of recursion and perform classification and regression tasks. Thus, they can perform a large number of tasks in a very simple way. They are used in healthcare to perform complex tasks like large data analysis, training models, and disease diagnosis.

Naive Bayes

Naive Bayes is a supervised learning algorithm that uses the Bayes theorem as its foundation. Although it is simple, it is highly effective for classification and filtering tasks. It is

used in healthcare to diagnose diseases based on symptoms, predict various future outcomes, and suggest the right medication for the patient.

Discriminant Analysis

This classification algorithm uses a discriminant function that helps separate the best classes from all the other available classes. It includes methods like Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA). LDA is most widely used as it involves a simple process compared to QDA. In healthcare, it is used to predict risks, pattern recognition in the patients, etc.

Random Forest

It is an ensemble-based supervised learning algorithm that constructs many decision trees during training and then outputs either classification—or regression-based results on each tree. Each tree acts as a different entity representing the different datasets. Its usage in healthcare includes identifying various insights from medical datasets, predicting future outcomes, etc.

K-Means Clustering

This is the most popular unsupervised learning algorithm in ML. Its main task is to separate datasets by identifying their similarities. K-means clustering is well known for data clustering, which is very important in healthcare. In healthcare, it is used for patient segmentation, disease classification, and medical image separation.

Q-Learning

Q-Learning is a reinforcement learning algorithm capable of making predictions based on its environment. It uses a parameter called Q-value, which indicates the possibility of success. Each step updates the Q-value to find the best results. These reinforcement algorithms can make decisions independently, which is very important in a few cases. It is used for treatment planning, optimizing healthcare operations, etc.

Deep Q-Networks (DQN)

DQN is another reinforcement learning algorithm. It is an enhanced version of the Q-learning algorithm, so-called deep Q-networks. It uses deep learning techniques, which makes it capable of working on the most complex datasets. In healthcare, it is used for drug discovery, decision-making, disease diagnosis, etc.

These are the most widely used Machine learning algorithms in healthcare. Using these algorithms enhances accuracy in test results, improving patient outcomes and making healthcare much safer. These algorithms can perform the most complex tasks. Thus, it reduces hectic tasks and burdens on healthcare experts [34–36].

Identified Gaps in Existing ML Technologies in Healthcare

Although the machine learning models have improved the healthcare diagnosis system, they still face few challenges in accurate predictions, data integration, and biased results especially in areas like breast cancer, lung cancer, and heart disease diagnosis.

In breast cancer screening, ML models show false positives at a rate of 10–15% according to studies. This means that many women undergo unnecessary biopsies and other follow-up procedures, which leads to an increase in healthcare costs and also affects patient's mental health. Along these there are a few cases where the false negatives are also shown by the models, this can lead to failure of the model to detect the disease which results in delaying of treatment and leads to fatality in some cases. This is the same in the case of Lung cancer diagnosis also where the models have shown an accuracy rate of around 75–80% only, failing in analysing the image data. These gaps in existing models can lead to overdiagnosis or missing in early-stage diagnosis ultimately affecting patient health outcomes.

Now considering heart disease diagnosis, using ML to predict diseases like myocardial infarction or arrhythmias has shown biased results because of training the model with certain demographic groups alone. For example, a study which is done in 2021 found that the predictive models for heart diseases are showing error rates up to 20% more in women compared to men. This is because the men are over-presented in the dataset which leads to biased results. This lack of generalizability in the dataset is the main cause of biased results.

The other challenge lies in integrating the machine learning model with existing medical records. A study in 2022 found that around 58% of healthcare institutions are struggling with interoperability where they are facing issues in integrating the patient's data from existing sources in EHRs (Electronic Health Records), Image scans, X-rays, and various other sources [37, 38].

How Prompt Engineering Addresses These Gaps

Prompt engineering overcomes these limitations of existing machine learning models in healthcare, particularly in cancer and heart disease diagnosis. Unlike the machine learning models which depend on specific datasets and lack generality

while providing results. Prompt engineering utilizes LLMs to process and interpret the data more effectively.

In breast cancer diagnosis, integrating prompt engineering can reduce false positives and negatives by up to 20% according to various studies. By refining the models with more patients historical and imaging data these models can perform much better. Now considering lung cancer, prompt engineering can improve interpretability. The existing models face challenges from high-dimensional data such as CT scans, where this subtle information can help in early disease diagnosis. So, incorporating the prompting models will guide the model to grab those crucial features that can be life-saving in a few cases.

In the case of heart disease diagnosis and treatment, prompt engineering can find out the bias and data integration problems that are present in the existing ML model. Because, unlike machine learning models, here we can tailor the prompts by considering the demographic information and the other features of the patient like EHRs, genetic information can be considered which can improve the accuracy by up to 15–20% compared to existing models. Thus, the integration of prompt engineering allows for real-time updates and continuous improvement of the model.

Thus, the incorporation of prompt engineering will ensure in providing of optimized treatment resulting in making healthcare better and more reliable for humans.

Future Directions

In the future, prompt engineering will be much more advanced than the present. Due to the rapid development of artificial intelligence, prompt engineering will be a mandatory skill for every individual. These are some of the aspects that will be largely focused on in the future, where prompt engineering will be effectively used [3, 18].

Integrating the Advanced Algorithms

Integrating advanced algorithms like deep learning and neural networks will make the prompt systems more accurate and precise. This ensures that the results being produced by the LLMs are trustable. AI and prompt engineering will be required in every domain in the future, not only in the healthcare field.

Integrating Wearables

Although it is being used on a small scale, integrating smart wearables is very important in the healthcare domain. It is crucial and can save lives in many situations for patients in remote locations. In the future, integrating smart wearables

with prompting techniques and AI will make healthcare much better and a lifesaver.

Expanding in Various Healthcare Domains

Prompt engineering is widely used in disease diagnosis and medication, so we can expand this into other healthcare domains in the future. Prompt engineering and AI can modernize the healthcare system in the future. There is much scope for using well-trained machines to perform complex surgeries with less intervention from doctors, which can be achieved by expanding the technology into various other domains.

Focusing on Ethical and Privacy Concerns

As prompt engineering is becoming common in healthcare, it is also very important to make sure the patient's details, such as personal information, test reports, and demographic details, are kept very secure, as data loss can be life-threatening for patients. This data should be handled very carefully as it is very sensitive information. The models must also be trained to follow basic ethical and privacy rules so that the patient can develop trust in technology.

Personalized Treatment

The critical aspect of prompt engineering will be its capability of prescribing medication for each patient based on their health condition. This ensures that every patient gets the right treatment and that their health condition improves. In the conventional method, doctors used to prescribe medication, but when replaced with machines, the task becomes very fast, and the results are more accurate.

Collaboration and Engagement

Concepts like collaboration and engagement play crucial roles in identifying the various advantages and challenges caused by technology. Global collaboration among top healthcare experts should be focused in the future. During this collaboration, they engage about the latest discoveries and research results they have found in their study; these results will be very useful in treating the patient [3, 18, 33].

Thus, these aspects should be focused on and implemented in the future to make the healthcare domain more advanced than all other domains.

Conclusion

Due to the vast advancement of Artificial Intelligence, prompt engineering has emerged as the latest trend.

This field of Prompt Engineering has great scope as this technology will be essential to handling AI models. Irrespective of domain, this technology will be crucial for every individual in the future as we are in the era of AI.

In conclusion, this paper has made significant contributions by demonstrating a range of prompt engineering techniques that can be applied to medical diagnostics. We have shown that prompt engineering not only improves diagnostic processes of diseases like breast cancer, lung cancer and heart conditions but it also fills the existing gaps in machine learning models thereby giving more refined outputs.

Our investigation was on how these strategies could be employed to improve the early detection and diagnosis of breast and lung cancers which are among the commonest types of cancers that cause many deaths across the world. Through prompt engineering, we will be able to improve accuracy in interpretation of mammography and CT scan results hence helping to identify malignant tumors at their earlier stages of development which is easier and less costly to treat. Additionally, this applies to heart conditions as well whereby prompt engineering methodologies facilitate with ECGs interpretations and other diagnostic tests thus making cardiac abnormalities diagnosis faster and more accurate.

The work highlights the prospects of prompt engineering to transform personalized medicine by enabling more customized diagnostic and treatment options. We can generate exact recommendations for treatment plans and follow-up care by integrating patient-specific data into prompt models, thus improving patient outcomes. In fast-paced clinical settings where timely access to accurate information is crucial, advanced techniques such as Tree of Thoughts and retrieval augmented generation enhance medical information retrieval accuracy and relevance.

Funding This work received no external funding.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. G.M. Muktadir, A Brief History of Prompt: Leveraging Language Models (Through Advanced Prompting). arXiv e-prints, arXiv:2310 (2023). <https://doi.org/10.13140/RG.2.2.12135.83364>
2. G. Marvin, N. Hellen, D. Jjingo, J. Nakatumba-Nabende, Prompt engineering in large language models. In *International Conference on Data Intelligence and Cognitive Informatics*, pp. 387–402. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-7962-2_30
3. A.S. George, A.H. George, The emergence of prompt engineering in India: assessing the potential for a new generation of AI

- talent. *Partners Univers. Int. Innov. J.* **1**(6), 1–18 (2023). <https://doi.org/10.5281/zenodo.10125681>
4. A. Kayid, *The Role of Artificial Intelligence in Future Technology* (Department of Computer Science The German University in Cairo, Cairo, 2020). <https://doi.org/10.13140/RG.2.2.12799.23201>
5. J. White, Q. Fu, S. Hays, M. Sandborn, C. Olea, H. Gilbert, et al., A prompt pattern catalog to enhance prompt engineering with chatgpt. *arXiv preprint arXiv:2302.11382*. (2023). <https://doi.org/10.48550/arXiv.2302.11382>
6. J. Wang, E. Shi, S. Yu, Z. Wu, C. Ma, H. Dai, et al., Prompt engineering for healthcare: Methodologies and applications. *arXiv preprint arXiv:2304.14670* (2023). <https://doi.org/10.48550/arXiv.2304.14670>
7. S. Ekin, Prompt engineering for ChatGPT: a quick guide to techniques, tips, and best practices. *Authorea Preprints* (2023). <https://doi.org/10.36227/techrxiv.22683919.v2>
8. C.S. Nag, E. Asif, Automated system for opinion detection of breathing problem discussions in medical forum using deep neural network. In *Proceedings of the 20th International Conference on Natural Language Processing (ICON)*, pp. 336–348 (2023).
9. G.S. Yadav, S.K. Dubey, An analytical approach to investigation of asthma diseases using soft computing techniques. In *Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2020*, Vol. 3, pp. 909–917. Springer, Singapore (2021). https://doi.org/10.1007/978-981-15-9774-9_83
10. K. Hari Priya, S. Alladi, S. Goje, M.N. Reddy, H. Nama, A Survey on Lung Cancer Detection and Location from CT Scan Using Image Segmentation and CNN. In *International Conference on Advanced Computing, Machine Learning, Robotics and Internet Technologies*, pp. 213–220. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-47224-4_19
11. U.M. Zamwar, K.N. Muneshwar, Epidemiology, types, causes, clinical presentation, diagnosis, and treatment of hypothyroidism. *Cureus* (2023). <https://doi.org/10.7759/cureus.46241>
12. M.S. Popoviciu, L. Paduraru, R.M. Nutas, A.M. Ujoc, G. Yahya, K. Metwally, S. Cavalu, Diabetes mellitus secondary to endocrine diseases: an update of diagnostic and treatment particularities. *Int. J. Mol. Sci.* **24**(16), 12676 (2023). <https://doi.org/10.3390/ijms241612676>
13. Z. Fekonja, R. Perkič, M. Strnad, S. Kmetec, B. McCormack, *Symptoms experienced by patients with acute myocardial infarction that the triage nurses should know in the emergency department: a systematic review* (Nurs. Care Educ. Res. Innov, 2023). <https://doi.org/10.1515/9783110786088-002>
14. J. Jamaluddin, M.S. Mohamed-Yassin, S.N. Jamil, M.A.M. Kamel, M.Y.A. Yusof, Frequency and predictors of inappropriate medication dosages for cardiovascular disease prevention in chronic kidney disease patients: a retrospective cross-sectional study in a Malaysian primary care clinic. *Heliyon* (2023). <https://doi.org/10.1016/j.heliyon.2023.e14998>
15. M.A. Mir, A.Y. Mir, Current treatment approaches to breast cancer. In *Therapeutic potential of cell cycle kinases in breast cancer*, pp. 23–51. Springer, Singapore (2023). https://doi.org/10.1007/978-981-19-8911-7_2
16. R. Lineros, L. Fernández-Delgado, A. Vega-Rioja, P. Chacón, B. Doukkali, J. Monteseirín, D. Ribas-Pérez, Associated factors of pneumonia in individuals with chronic obstructive pulmonary disease (COPD) apart from the use of inhaled corticosteroids. *Biomedicine* **11**(5), 1243 (2023). <https://doi.org/10.3390/biomedicine11051243>
17. L.A. Huppert, O. Gumusay, D. Idossa, H.S. Rugo, Systemic therapy for hormone receptor-positive/human epidermal growth factor receptor 2-negative early stage and metastatic breast cancer. *CA Cancer J Clin* **73**(5), 480–515 (2023). <https://doi.org/10.3322/caac.21777>
18. B. Meskó, Prompt engineering as an important emerging skill for medical professionals: tutorial. *J. Med. Internet Res.* **25**, e50638 (2023). <https://doi.org/10.2196/50638>
19. B. Chen, Z. Zhang, N. Langrené, S. Zhu, Unleashing the potential of prompt engineering in large language models: a comprehensive review. *arXiv preprint arXiv:2310.14735* (2023). <https://doi.org/10.48550/arXiv.2310.14735>
20. D.F. Sittig, A. Wright, E. Coiera, F. Magrabi, R. Ratwani, D.W. Bates, H. Singh, Current challenges in health information technology-related patient safety. *Health Inform. J.* **26**(1), 181–189 (2020). <https://doi.org/10.1177/1460458218814893>
21. Y. Lamo, F. Rabbi, R. Barendregt, Challenges for technology innovation in health care. In *2016 International Conference on Advances in Electrical, Electronic and Systems Engineering (ICAEEES)*, pp. 48–53. IEEE (2016). <https://doi.org/10.1109/ICAEEES.2016.7888007>
22. B.C. Stahl, B.C. Stahl, Ethical issues of AI. *Artificial Intelligence for a better future: an ecosystem perspective on the ethics of AI and emerging digital technologies*, pp. 35–53 (2021). https://doi.org/10.1007/978-3-030-69978-9_4
23. J.T. Loud, J. Murphy, Cancer screening and early detection in the 21st century. In *Seminars in oncology nursing*, Vol. 33, No. 2, pp. 121–128. WB Saunders (2017). <https://doi.org/10.1016/j.soncn.2017.02.002>
24. H.S. Choi, J.Y. Song, K.H. Shin, J.H. Chang, B.S. Jang, Developing prompts from large language model for extracting clinical information from pathology and ultrasound reports in breast cancer. *Radiat. Oncol. J.* **41**(3), 209 (2023). <https://doi.org/10.3857/roj.2023.00633>
25. A. Elfink, I. Vagliano, A. Abu-Hanna, I. Calixto, Soft-prompt tuning to predict lung cancer using primary care free-text Dutch medical notes. In *International Conference on Artificial Intelligence in Medicine*, pp. 193–198. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-34344-5_23
26. D. Amicizia, M.F. Piazza, F. Marchini, M. Astengo, F. Grammatico, A. Battaglini, et al. Systematic review of lung cancer screening: advancements and strategies for implementation. In *Healthcare*, Vol. 11, No. 14, p. 2085. MDPI (2023). <https://doi.org/10.3390/healthcare11142085>
27. Y. Muhammad, M. Tahir, M. Hayat, K.T. Chong, Early and accurate detection and diagnosis of heart disease using intelligent computational model. *Sci. Rep.* **10**(1), 19747 (2020). <https://doi.org/10.1038/s41598-020-76635-9>
28. P. Gayathri, N. Jaisankar, Comprehensive study of heart disease diagnosis using data mining and soft computing techniques. *Int. J. Eng. Technol.* **5**(3), 2947–2958 (2013)
29. A. Sharma, R.A. Harrington, M.B. McClellan, M.P. Turakhia, Z.J. Eapen, S. Steinhilb et al., Using digital health technology to better generate evidence and deliver evidence-based care. *J. Am. Collage Cardiol.* **71**(23), 2680–2690 (2018)
30. H. Thimbleby, Technology and the future of healthcare. *J Public Health Res* **2**(3), 2013 (2013). <https://doi.org/10.4081/jphr.2013.e28>
31. B. Khan, H. Fatima, A. Qureshi, S. Kumar, A. Hanan, J. Hussain, S. Abdullah, Drawbacks of artificial intelligence and their potential solutions in the healthcare sector. *Biomed. Mater. Dev.* **1**(2), 731–738 (2023). <https://doi.org/10.1007/s44174-023-00063-2>
32. Y.K. Alotaibi, F. Federico, The impact of health information technology on patient safety. *Saudi Med. J.* **38**(12), 1173 (2017). <https://doi.org/10.15537/smj.2017.12.20631>
33. J. Bajwa, U. Munir, A. Nori, B. Williams, Artificial intelligence in healthcare: transforming the practice of medicine. *Future Healthc J* **8**(2), e188 (2021). <https://doi.org/10.7861/fhj.2021-0095>
34. R.B. Sathvik, T.S.P.R. Kumar, C.V.N. Rugvidh, V. Rupesh, C.R. Mohan, N. Subramanyan, S. Kiran, A.A. Kumar, Machine learning in healthcare technology: insights, challenges, roles and

- applications. Biosci. Biotech. Res. Commun. (2023). <https://doi.org/10.21786/bbrc/16.3.8>
35. M. Piri, Review of supervised learning in healthcare. In: *5th International Congress on Engineering, Technology & Innovation* (2023).
 36. Y. Wang, Y. Zhao, T.M. Therneau, E.J. Atkinson, A.P. Tafti, N. Zhang et al., Unsupervised machine learning for the discovery of latent disease clusters and patient subgroups using electronic health records. *J. Biomed. Inform.* **102**, 103364 (2020). <https://doi.org/10.1016/j.jbi.2019.103364>
 37. C.C.L. Dos Santos, A.S. Matharoo, E.P. Cueva, U. Amin, A.A.P. Ramos, N.K. Mann et al., The influence of sex, age, and race on coronary artery disease: a narrative review. *Cureus* **15**, 10 (2023). <https://doi.org/10.7759/cureus.47799>
 38. G. Chugh, S. Kumar, N. Singh, Survey on machine learning and deep learning applications in breast cancer diagnosis. *Cogn. Comput.* **13**(6), 1451–1470 (2021). <https://doi.org/10.1007/s12559-020-09813-6>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.